James “Jake” Gearon — Curriculum Vitae

Location	Chapel Hill, NC, USA	**Email**	jake.gearon@gmail.com
Citizenship	USA	**Website**	sandfrom.space
		Profiles	

Education

2021–2025 Ph.D in Sedimentology & Remote Sensing - Indiana University–Bloomington

Advisor: Douglas Edmonds
GPA: 3.8 / 4.0
Ph.D Dissertation *Investigating River Avulsion Dynamics with Large-Scale Remote Sensing Data Operations*

2019–2021 MS in Geology - The University of Texas at Austin

Advisors: Cornel Olariu & Ronald J. Steel
GPA: 4.0 / 4.0
Masters Thesis - *The Supply-Generated Sequence: A New Sequence–Stratigraphic Model for Fluvio-Lacustrine Deposits*

2015–2019 BS Honors, Special Honors in Geology - The University of Texas at Austin

Certificate in Creative Writing
GPA: 3.6 / 4.0
Undergraduate Thesis - *Geomorphic Influences on Canopy Architecture of Larrea tridentata in Northern Mojave Desert, Nevada, USA*
Advisor: Michael H. Young

Publications

[1] **James H. Gearon** and Douglas A. Edmonds. “Alluvial ridge morphology on avulsing rivers”. In: *Sedimentology* 73.6 (July 2026), pp. 2311–2337. DOI: 10.1111/sed.70134.

[2] E. A. Barefoot, **J. H. Gearon**, and D. A. Edmonds. “O Levee, Where Art Thou? Measuring the Abundance of Natural River Levees Across the Contiguous USA”. In: *Journal of Geophysical Research: Earth Surface* 131.4 (2026), e2025JF008642. DOI: 10.1029/2025JF008642.

[3] **James H. Gearon** and Douglas A. Edmonds. “River Avulsion Precursors Encoded in Alluvial Ridge Geometry”. In: *Geophysical Research Letters* 52.8 (Apr. 2025), e2024GL114047. DOI: 10.1029/2024GL114047.

[4] Aman Agarwal, **James Gearon**, Raksha Rank, and Etienne Chenevert. *Fighting Fires from Space: Leveraging Vision Transformers for Enhanced Wildfire Detection and Characterization*. Apr. 2025. DOI: 10.48550/arXiv.2504.13776. arXiv: 2504.13776 [cs].

[5] Joseph S. Levy, Thomas F. Subak, Ian Armstrong, Izzy King, Lingfeng Kuang, Lily Kuentz, **James H. Gearon**, Sophie Naylor, M.C. Rapoza, and Haobo Wang. "Martian Chaos terrain fracture geometry indicates drainage and compaction of laterally heterogeneous confined aquifers". In: *Icarus* 426 (2025), p. 116377. DOI: 10.1016/j.icarus.2024.116377.

[6] **James H. Gearon**, Harrison K. Martin, Clarke DeLisle, Eric A. Barefoot, David Mohrig, Chris Paola, and Douglas A. Edmonds. "Rules of river avulsion change downstream". In: *Nature* 634.8032 (Sept. 2024), pp. 91–95. DOI: 10.1038/s41586-024-07964-2.

[7] T. H. Doane, **J. H. Gearon**, H. K. Martin, B. J. Yanites, and D. A. Edmonds. "Topographic Roughness as an Emergent Property of Geomorphic Processes and Events". In: *AGU Advances* 5.5 (Sept. 2024), e2024AV001264. DOI: 10.1029/2024AV001264.

[8] Puyu Liu, Chenglin Gong, **James H. Gearon**, Dayong Guan, Qiming Wang, Kun Qi, and Dongwei Li. "Increased sediment connectivity between deltas and deep-water fans in closed lake basins: A case study from Bozhong Sag, Bohai Bay Basin, China". In: *Sedimentary Geology* 460 (Feb. 2024), p. 106561. DOI: 10.1016/j.sedgeo.2023.106561.

[9] **James H. Gearon**, Cornel Olariu, and Ronald J. Steel. "The Supply-Generated Sequence: A Unified Sequence-Stratigraphic Model for Closed Lacustrine Sedimentary Basins with Evidence from the Green River Formation, Uinta Basin, Utah, U.S.A." In: *Journal of Sedimentary Research* 92.9 (Sept. 2022), pp. 813–835. DOI: 10.2110/jsr.2021.096.

[10] **James H Gearon** and Michael H Young. "Geomorphic controls on shrub canopy volume and spacing of creosote bush in northern Mojave Desert, USA". In: *Landscape Ecology* 36.2 (Feb. 2021), pp. 527–547. DOI: 10.1007/s10980-020-01149-8.

Data Products

[1] **James H. Gearon**, Elizabeth H. Altenau, Tamlin M. Pavelsky, Michael T. Durand, Niek Collot d'Escury, Xiao Yang, Pierre-Olivier Malaterre, Renato P. d. M. Frasson, and Liam Bendezu. *SWOT River Database (SWORD), version 17c*. Version v17c. Zenodo, July 2026. DOI: 10.5281/zenodo.21415370.

Research Experience

Aug. 2025 – Present Global Hydrology Lab, University of North Carolina, Chapel Hill
Post-Doctoral Researcher

Working with Dr. Tamlin Pavelsky on the Surface Water and Ocean Topography (SWOT) satellite mission, supported by NASA SWOT Science Team funding. The central project makes the first global, satellite-based measurements of hydraulic head difference across 20,546 river obstructions, from low-head barriers to large dams. Barrier inventories record where obstructions are, not how much each changes the river; these measurements show how much connectivity removal would return and where restoration spending does the most. A second project measures seasonal backwater dynamics on the world's large coastal rivers.

2025-Present SWOT River Database (SWORD)
Lead Developer and Maintainer

I lead development of SWORD, the global river hydrography dataset that defines the reaches and nodes on which the SWOT mission's river data products are built: 248,673 reaches and 11.1 million nodes across six continental regions. I am first author on the most recent public release, version 17c (2026), which brought SWOT-measured water-surface elevation, width, and slope into the database for the first time. The role covers scientific design, the processing and validation pipeline, versioned public release, and support for an international user community.

Sep. 2023-Sep. 2024 Army Engineer Research and Development Center (ERDC) Grant
Research Assistant

Worked as part of a multi-institution research group focused on automated inspection of levee systems in the United States. My role consisted of building a system architecture to compare elevation profiles recorded in the National Levee Database (NLD) with USGS 3DEP 1m lidar-derived DEMs.

Aug. 2021-May. 2025 Indiana University Dept. of Earth and Atmospheric Science
Doctoral Student/Candidate

Advised by Dr. Douglas Edmonds, Committee Members: Dr. Brian Yanites, Dr. Ben Kravitz, Dr. Santo Fortunato (Minor Advisor), and Dr. Chris Paola (External)

Feb. 2020-Jan. 2021 Earth Science Information Partners (ESIP) Winter 2020 RFP Grant
Principal Investigator

Authored a proposal as PI to develop the Global Lake Level Database. Three separate databases (USGS, G-REALM, and HydroWeb) were merged and are hosted in an AWS RDS Database with an attached API. To provide ease-of-access to endusers, I developed a pythonic front-end LakePy designed to rapidly access up-to-date water levels from the Global Lake Level Database back-end.

Aug. 2019-May 2021 UT Austin, Jackson School of Geosciences, Dynamic Stratigraphy Group
Master's Student

Advised by Drs. Ron Steel and Cornel Olariu, my third committee member was Dr. Brian Horton. Developed thesis project on constraining single lacustrine sedimentation "events" or deposodes and how they appear in the rock record using drone orthophotogrammetry, logged section and well-log analysis. My field area was in Uinta Basin, UT in the Middle Green River formation. To better understand the high-frequency response of lacustrine deltas to base level rise, I also investigated modern lake deltas using original remote sensing algorithms.

June 2017- Bureau of Economic Geology, Undergraduate Honors Thesis
May 2019 *Undergraduate Research Assistant to Dr. Michael Young*

Built algorithms in Python to measure and chart spatial properties of vegetation of a large alluvial deposit in Southern California based on remote sensing data, presented poster at AGU 2018 and GSA 2019. Gained experience in processing and analyzing Lidar and Multi-Spectral Imagery data using Python. Wrote undergraduate thesis based on this work finding statistically significant relationships between plant size and spacing based on soil age and clay content.

Nov. 2015- University of Texas Institute for Geophysics, Undergraduate Research Project
April 2017 *Undergraduate Research Assistant to Drs. Joseph Levy & Tim Goudge*

Assisted in the mapping and general geomorphological analysis of Martian Chaos terrain, utilizing ArcGIS and Python for detailed geostatistical evaluation and interpretation of data. Presented poster at 2017 Lunar & Planetary Science Conference in Houston, Texas

Field Experience

June 2024 NSF 'Rivers of the Andes Field Trip' Course

The RAFT program provided a unique opportunity to engage in transdisciplinary and cross-cultural collaboration with students and researchers from diverse backgrounds and expertise. This experience in Colombia highlighted the importance of international collaboration and knowledge transfer in advancing our understanding of complex environmental challenges.

May 2023 IU Colorado Plateau Class

Co-led a week-long trip to Utah as part of a class for undergraduates and graduate students on the structure and stratigraphy of the Colorado Plateau with Dr. Kaj Johnson. Lessons covered sequence stratigraphy, depositional environments, sedimentary data collection, log interpretation, and general field skills. *Takeaways: teaching experience, leading a group of students with varying geoscience knowledge*

July 2020 Field Season in 9-Mile Canyon, Uinta Basin, Utah

Conducted a month of field-work in the Sunnyside Delta Interval (after Remy et al., 1991) of the Middle Green River Formation. 1000m of logged section was produced, as well as 20GB of Drone Imagery. *Takeaways: lacustrine sequence stratigraphy, self-directed field work, drone imagery acquisition*

Nov. 2019 Borrego Springs Field Trip w/ Ron Steel & Cornel Olariu

Attended trip for "Clastic Depositional Systems". Logged 100m of section in the Lycium turbidite units. *Takeaways: rift-basin sequences, distinguishing source from turbidite grain character, antidune outcrop examples*

Mar. 2019 Peruvian Andean Geology Trip w/ Dan Breecker & Liz Cassel

Attended 10-day trip as a part of the UT Austin Undergraduate Research Honors Program. *Takeaways: Andean tectono-stratigraphy, Fold-Thrust-Belt sedimentology/sequences, growth strata*

July 2018 UT Austin Field Camp

Attended the 6-week, 6,600-mile mobile field course from UT covering many aspects of geological study. Professors included David Mohrig, Tim Goudge, Rowan Martindale, Dan Breecker, Jaime Barnes, Danny Stockli, Mark Helper. *Takeaways: geologic field mapping, rock/mineral ID, stratigraphic log interpretation, and more.*

Mar. 2018 Book Cliffs Sedimentology Short Course w/ Ron Steel

Attended a 10-day trip to NE Utah as a part of the UT Austin Undergraduate Honors Research Program. *Takeaways: western interior seaway formation/deposition, hyperpicnites, lacustrine carbonates, shelf behavior primer*

Teaching Experience

May 2023 Co-Instructor: Colorado Plateau Field Course

Co-led a week-long trip to Utah as part of a class for undergraduates and graduate students on the structure and stratigraphy of the Colorado Plateau with Dr. Kaj Johnson. Lessons covered sequence stratigraphy, depositional environments, sedimentary data collection, log interpretation, and general field skills

Fall 2020 Teaching Assistant: GEO 416K *Earth Materials*

Course taught by Dr. Elizabeth Catlos. I led two 4-hour labs and instructed students on a range of topics from mineralogy, crystallography, and petrology and assisted professor in re-designing course for online instruction due to COVID-19

Spr. 2020 Teaching Assistant: GEO 316P *Sedimentary Rocks*

Course taught by Dr. Cornel Olariu. I Created quizzes and extra credit assignments for 84 petroleum engineering students while achieving an average student evaluation score of 4.3/5.00. Further assisted professor in re-designing course for online instruction due to COVID-19

Fall 2019 Teaching Assistant: GEO 401 *Physical Geology*

Course taught by Drs. Jaime Barnes & Daniel Breecker. I led three two-hour lab sessions with 45 students total emphasizing practical aspects of geology including mineral & rock identification, field hydrology, map-reading, and structural interpretation. Achieved an average student evaluation score of 4.63/5.00

Research Mentoring

Undergraduate researchers, University of North Carolina at Chapel Hill, 2025–present

- **Lilian Soyke**: undergraduate thesis testing whether gradually varied flow theory, linearized about normal depth, predicts the upstream excursion of storm surge in the river water surface
- **Maggie Kiley**: storm database and skew surge calculations for the same project; co-mentored with Dr. Antonia Sebastian
- **Corey Reaves**: basin-scale controls on avulsion occurrence; co-advised with Dr. Sarah George
- **Catherine Quayle**: surface water–groundwater connection in the Nebraska Sandhills from SWOT lake heights paired with lake and well gauges, and its bearing on Ogallala Aquifer recharge. First-author poster, AGU 2026
- **Emma Johnson**: undergraduate thesis on how reliably SWOT detects recorded floods and their magnitude; manuscript in preparation
- **Katie Fear**: undergraduate thesis harmonizing existing flood inventories into a global database, with an LLM pipeline that extracts floods from news articles; found that flood reporting is biased along socioeconomic lines
- **Caroline Lorio**: global QA/QC of lake flags in SWORD
- **Natalie Williams**: QA/QC of coastal river mainstem routing in SWORD
- **Jesse Williams**: lake–groundwater connectivity in Florida karst terrain

Employment History

May 2019- Chevron Corporation ETC, 1111 Bagby St. Houston, TX 77002
Aug. 2019 *Data Engineering Intern*

Developed a Python package for full stack analytics model deployment from DevOps repository to production-level, server-hosted models to be queried for prediction results via REST API. Built in customizable functionalities to track model metrics throughout successive runs and fully deploy most accurate models utilizing container services. Gained experience in Software Engineering, VCS, Spark Python, Microsoft Azure, Databricks, and MLflow.

July 2018 - Chevron Corporation ETC, 1111 Bagby St. Houston, TX 77002
Aug. 2018 *Technical Computing Intern*

- Worked in Subsurface Workflows dept. on workflow regression testing for Petrel 2016.3, 2017.4, and 2018.1.
- Worked with Petrophysics team on machine learning applications for lithofacies identification. Gained experience in Microsoft Visual Studio, Geolog 18.0, and Pythonic data analytics.

Service Activities

Fall 2021 - Present Peer Review
Manuscripts and Proposals

Journals (20 reviews): Nature Communications, Geology, Geophysical Research Letters, Water Resources Research, Journal of Geophysical Research: Earth Surface, Sedimentology, Earth Surface Dynamics, Geomorphology, Quaternary Research, Basin Research, Journal of Sedimentary Research, PNAS Nexus, Sedimentary Geology, Marine and Petroleum Geology, Texas Water Journal, Journal of Open Source Software

Proposals: National Science Foundation, Directorate for Geosciences (ad hoc)

Fall 2022 - Spr. 2023 Earth and Atmospheric Science Graduate Student Association
Faculty Representative

Function was to meet with Dept. Chair and Director of Graduate Studies monthly to relay information from the graduate student population as well as to get updates on departmental goings-on. Additionally served on the Faculty Hiring Committee.

Fall 2020 - Spr. 2021 UT Austin: Technology Enhanced Education Oversight Committee (C-14)
Graduate Student Assembly Representative

Function: to evaluate and formulate policy on technology-enhanced education and make recommendations on such matters to the Faculty Council. We meet monthly to discuss current issues within the context of technology and higher education. This year (2021) we have passed a resolution recommending the creation of a school-wide dedicated Education Technology Officer to better coordinate between stakeholders.

Media Coverage

Oct. 2024 What leads rivers to suddenly change course? *ScienceNews*

Oct. 2024 Scientists discover new way of predicting where and when rivers suddenly change course and flood *National Science Foundation*

Sep. 2024 Breakthrough study from IU Scientists predicts catastrophic river shifts that threaten millions worldwide *UN Office for Disaster Risk Reduction*

Dec. 2020 'Big Data' Enables First Census of Desert Shrub *Jackson School of Geosciences*

Grants & Fellowships

Sep. 2024- Indiana University Dissertation Year Fellowship
May 2025 *Ph.D Candidate*

Awarded by the Indiana University College of Arts and Sciences on a yearly basis

Feb. 2020- Earth Science Information Partners (ESIP) Winter RFP 2020
Dec. 2020 *Principal Investigator*

Funded $8,000: Developing an Open-Source Workflow and Toolset for Quantifying Lacustrine Sedimentation using Publicly Available Data

Aug. 2021- Grassman Fellowship
May. 2022 *Ph.D Student*

Awarded to one candidate by the Indiana University Department of Earth & Atmospheric Sciences on a yearly basis

Presentations & Posters

Invited

Department Colloquium: UNC Chapel Hill, Earth, Marine & Environmental Sciences

- **Sept. 2026**: Reading Rivers from the Rock Record to the Water Surface

Department of Geography: Oxford University

- **June. 2026**: Visualizing Coastal Backwater Dynamics with SWOT

Landscapes Live Seminar (Online)

- **March. 2025**: How River Avulsions Work

University of Delaware Colloquium

- **March. 2025**: How River Avulsions Work

EEPS Brown Bag: Washington University in St. Louis (WashU)

- **Feb. 2025**: How River Avulsions Work

Smith Lecture: University of Michigan Earth & Environmental Sciences Dept.

- **Feb. 2025**: How River Avulsions Work

USGS CDI Geomorphology Group

- **Oct. 2024**: Observing River Avulsion from Space

USGS CDI Information Technology & Interpretability Group

- **Sep. 2021**: LakePy: A database and API for lake water level analysis

Water, Climate & Earth Seminar (UT)

- **Oct. 2020**: Geomorphic controls on shrub canopy volume and spacing of creosote bush in northern Mojave Desert, USA

Eco-Lunch (UT)

- **Oct. 2020**: Geomorphic controls on shrub canopy volume and spacing of creosote bush in northern Mojave Desert, USA

Contributed

[1] **J. H. Gearon** and T. Pavelsky. "SWOT Reveals the Hydraulic Burden of River Obstructions Worldwide". In: *AGU Fall Meeting 2026*. AGU. Dec. 2026.

[2] C. Quayle, **J. H. Gearon**, and E. Levenson. "Ogallala Surface-Groundwater Interactions Observed by SWOT in Sandhills, NE, USA". In: *AGU Fall Meeting 2026*. AGU. Dec. 2026.

[3] **J. H. Gearon**, T. Pavelsky, and J. Wang. "SWOT Reveals a Spectrum of Seasonal Backwater Dynamics on Large, Coastal Rivers". In: *AGU Fall Meeting 2025*. AGU. Dec. 2025.

[4] **James H. Gearon** and Douglas A. Edmonds. "River Avulsion Precursors Encoded in Alluvial Ridge Geometry". In: *RCEM 2025, Barcelona*. 2025.

[5] Eric Alexander Barefoot, **James Hooker Gearon**, JeongYeon Han, Tamlin Pavelsky, and Douglas A Edmonds. "Measuring the Abundance and Dimensions of Natural River Levees Across the Contiguous USA with 3DEP Lidar". In: *AGU Fall Meeting 2024*. 2024.

[6] **James Hooker Gearon** and Douglas A Edmonds. "Avulsion Risk Encoded in Channel Topography". In: *AGU Fall Meeting 2024*. 2024.

[7] Yuan Li, **James Hooker Gearon**, Arya Mimi Gotoh, and Douglas A Edmonds. "Assessing the Degradation of Artificial Levees across the US with Remotely Sensed Data". In: *AGU Fall Meeting 2024*. 2024.

[8] Cheryl Lyn Watts, Luke Jospeh Gezovich, Sam Bagge, Alejandro Arboleda, and **James Hooker Gearon**. "Determining Imminent Avulsion Risk on the Nechi River, Colombia". In: *AGU Fall Meeting 2024*. 2024.

[9] Eric Barefoot, **James Hooker Gearon**, Jeongyeon Han, and Douglas A Edmonds. "Measuring the abundance of natural river levees across the contiguous USA with 3DEP lidar". In: *AGU Fall Meeting 2023*. 2023.

[10] Tyler Doane, Aman Agarwal, **James Hooker Gearon**, Brian J Yanites, and Douglas A Edmonds. "Automated Mapping of Tree Uprooting in Lidar Data at Scale Using Deep Learning". In: *AGU Fall Meeting 2023*. 2023.

[11] Tyler Doane, **James Hooker Gearon**, Harrison Martin, Brian J Yanites, and Douglas A Edmonds. "Generalized Theory for Relating Topographic Roughness to Ecologic, Hydrologic, and Atmospheric Processes". In: *AGU Fall Meeting 2023*. 2023.

[12] **James H. Gearon** and Douglas A. Edmonds. "Investigating Hypotheses for River Avulsion using Space-Borne Lidar". In: *International Conference of Fluvial Sedimentology*. 2023.

[13] **James H. Gearon**, Harrison K. Martin, Clarke DeLisle, Eric A. Barefoot, David Mohrig, Chris Paola, and Douglas A. Edmonds. "Spatial Variability of Avulsion Initiation Mechanisms". In: *Rivers, Coastal, and Estuarine Morphodynamics Conference*. 2023.

[14] **James Hooker Gearon**, Eric Barefoot, Tamlin Pavelsky, and Douglas A Edmonds. "Characterizing vertical relief in the world's riparian zones". In: *AGU Fall Meeting 2023*. 2023.

[15] **James H. Gearon** and Douglas A. Edmonds. "Space Lasers for Rivers: Using ICESat-2 Data for Fluvial Geomorphology". In: *ESIP Winter Meeting 2022*. ESIP. Jan. 2022.

[16] Connor Broaddus, **James H. Gearon**, Douglas A Edmonds, and Efi Foufoula-Georgiou. "Topographic Scaling Reflects Process Differences on Alluvial and Fluvial Fans". In: *AGU Fall Meeting 2022*. Vol. 2022. 2022, EP36A–08.

[17] Etienne Chenevert, **James H. Gearon**, and Douglas A Edmonds. "Using Explainable Machine Learning to Investigate the Controls of Vertical Accretion on the Mississippi River Deltaic Plain, Louisiana, USA." In: *AGU Fall Meeting 2022*. Vol. 2022. 2022, EP52D–0786.

[18] **James H. Gearon** and Douglas A. Edmonds. "Investigating Hypotheses for River Avulsion using Space-Borne Lidar". In: *AGU Fall Meeting 2022*. AGU. 2022.

[19] **James H Gearon**. "LakePy: A Python Package for Accessing and Manipulating Lacustrine Time-Series Data". In: *USGS & ESIP IT&I & CDI TechStack Working-group*. USGS & ESIP. 2021.

[20] **James H Gearon**, Cornel Olariu, and Ronald J Steel. "The Supply-Generated Sequence: A New Sequence-Stratigraphic Model for Fluvio-Lacustrine Deposits". In: *IMAGE Online Meeting, USA-2021*. IMAGE. 2021.

[21] **James H. Gearon** and Douglas A. Edmonds. "Automated River Profiling from Space: A 21st Century Solution to an Inconvenient Problem". In: *AGU Fall Meeting 2021*. AGU. 2021.

[22] **James (Jake) Gearon** and John Franey. "Developing an Open-Source Database and Pythonic Toolset for Quantifying Lacustrine Sedimentation using Publicly Available Data". In: *ESIP Summer Meeting 2020*. ESIP. July 2020. DOI: 10.6084/m9.figshare.12628511.v1.

[23] **James H Gearon**, Cornel Olariu, and Ronald J Steel. "Proximal to Distal Changes of Thin Lacustrine Deltas in the Ancient and Modern". In: *AAPG Annual Online Meeting, USA-2020*. AAPG. 2020.

[24] **James Hooker Gearon**, Cornel Olariu, and Ronald Steel. "Quantifying High-Frequency Lacustrine Sedimentation Patterns using Publicly Available Data". In: *AGU Fall Meeting 2020*. AGU. 2020.

[25] Michael Young and **James Hooker Gearon**. "Geomorphic Controls on Shrub Canopy Volume and Spacing of Creosote Bush in Northern Mojave Desert, USA". In: *AGU Fall Meeting 2020*. AGU. 2020.

[26] **James H Gearon**, Cornel Olariu, and Ronald J Steel. "Unveiling lake delta architectures in modern and ancient systems by mapping river flood deposodes". In: *RioMAR Annual Meeting in Houston, Texas, USA-2019*. RioMAR. 2019.

[27] **James H Gearon** and Michael H Young. "Geomorphic Controls on Shrub Canopy Size and Spacing of Creosote Bush in Northern Mojave Desert, USA". In: *GSA Annual Meeting in Phoenix, Arizona, USA-2019*. GSA. 2019.

[28] **James Hooker Gearon** and Michael Young. "Geomorphic Influence on Canopy Structure in Eldorado Valley, Nevada". In: *AGU Fall Meeting 2018*. AGU. 2018, H13G–1799.

[29] **James H Gearon**, Joseph S Levy, and Timothy A Goudge. "Making Sense of Chaos: Geomorphic Investigations of Martian Chaos Terrain". In: *LPSC Annual Meeting in Houston, Texas, USA-2017*. LPSC. 2017.

Software

- **chronozarr**: Zarr v3 format and browser viewer for raster time series hosted as static files. Anyone with the link can scrub through years of satellite imagery and query stored pixel values with no tile server; released on PyPI and npm
- **dgov**: orchestration kernel for multi-agent AI coding. Dispatches agents into isolated git worktrees and compiles plans to DAGs that run through dispatch, review, merge, and evaluation
- **MCP servers**: firepass-mcp (Kimi via Fireworks AI as an agentic coding assistant with a full tool loop); gemma-mcp (the same interface on a locally hosted Gemma 4 via MLX); contributor to qgis_mcp
- **llm-infra**: self-hosted llama.cpp inference on the lab's River compute server (2× NVIDIA L40S)
- **walker**: autonomous research agent; 189+ experiments on exploration–exploitation in deceptive graph routing
- **quarry**: geospatial execution substrate with typed contracts from ingest to output and a registry that records lineage
- **topos**: TypeScript monorepo for geospatial visualization (interactive globes with satellite tracking), including the geo-renderer MCP server for D3 map generation and GDAL processing
- **BASED**: XGBoost estimator of river channel depth from width, slope, and discharge, served as an API and Streamlit app
- **drsedman**: Gemini-based structured extraction from geomorphology and sedimentology papers
- **LakePy**: Python front end to the Global Lake Level Database

Honors & Awards

Oct. 2024 AGU EPSP Early Career Spotlight

Nov. 2020 Invited Speech: Jackson School of Geosciences Scholarship Luncheon

May 2019 - 7 Semesters of University Honors

May 2019 - 5 Semesters Jackson Matrix Scholarship

April 2017-2nd Place in R.L. Folk and E. F. McBride Petrography Contest

Professional Organizations

American Association of Petroleum Geologists - *Active Member*

Geological Society of America - *Active Member*

The Society of Sedimentary Geology - *Active Member*

Earth Science Information Partners - *Project Principal Investigator 2020*

EarthCube - *Member*

Programming & Software

- **Proficient in:**

  Python - Pandas, NumPy, scikit-learn, Rasterio, GDAL, SciPy, Flask, GeoPandas, Boto3, Dask, Xarray, DuckDB, OpenCV, Earth Engine, STAC

  SQL - PostgreSQL, PostGIS, DuckDB SQL

  AI Engineering - agent harnesses and tool loops, MCP server development, multi-agent orchestration, fine-tuning Gemma and Qwen models on geoscientific knowledge from paper embeddings, local model serving (MLX, llama.cpp), RAG pipelines, Claude, Gemini, and OpenAI APIs, Claude Code, Codex

- **Also:**

  Julia - numerical computing
  JavaScript - web development
  Cloud & DevOps - Docker, Poetry, uv, CI/CD, Git
  AWS - RDS, S3, Aurora Serverless, Lambda, API Gateway
  GIS Software - ArcGIS, QGIS, GeoLibre, Global Mapper
  IDE & Development - PyCharm, VSCode, Jupyter
  Statistical Analysis - Spatial Statistics, GLMs, LASSO/RIDGE